

A tray has 4 rows of 5 cups.

Use repeated addition to find the number of cups.

Cups: _____

True or false?

3 rows of 5 tiles is $5 + 5 + 5 = 15$ tiles.

A. True

B. False

Which addition gives the number of squares in 2 rows of 4?

A. $2 + 2 + 4$

B. $4 + 4$

C. $2 + 4$

D. $4 + 2$

How many tiles are in 3 rows of 3?

Tiles: _____

True or false?

4 rows of 4 tiles is $4 + 4 = 16$ tiles.

A. True

B. False

SET 52 • 3.MD.C.7.b

FORWARD 1

Use the area formula to solve real-world problems with larger numbers • Grade 4 • Measurement & Data • 4.MD.A.3

A patio is 12 feet by 11 feet.

What is its area?

Area: _____

True or false?

A 15-m by 20-m field has an area of 300 square meters.

A. True

B. False

A wall is 9 ft tall and 14 ft wide.

What is its area?

- A.** 126 square feet
- B.** 46 square feet
- C.** 23 square feet
- D.** 116 square feet

A pool is 25 m long and 10 m wide.

What is its area?

Area: _____

True or false?

A 13-cm by 7-cm card has an area of 81 square cm.

A. True

B. False

SET 52 • 3.MD.C.7.b

FORWARD 2

Find the area of a rectangle with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.b

Find the area of a rectangle that is $\frac{2}{3}$ m by $\frac{3}{4}$ m.

Area: _____

True or false?

A rectangle $2\frac{1}{2}$ ft by 2 ft has an area of 5 square feet.

- A. True
- B. False

A square has sides that are $\frac{1}{3}$ yard long.

What is its area?

A. $\frac{1}{6}$ square yard

B. $\frac{4}{3}$ square yards

C. $\frac{2}{3}$ square yard

D. $\frac{1}{9}$ square yard

A rug is 4 feet by $1\frac{1}{2}$ feet.

What is its area?

Area: _____

True or false?

A rectangle $\frac{1}{2}$ inch by $\frac{1}{4}$ inch has an area of $\frac{3}{4}$ square inch.

- A. True
- B. False

Use tiling to show that the area of a rectangle with sides a and $b + c$ is $a \times b + a \times c$

MAIN	Grade 3 main question	3.MD.C.7.c
BACKWARD 1	Split an array into two parts and add	2.OA.C.4
BACKWARD 2	Add tens and ones within 100	2.NBT.B.5
FORWARD 1	Use area models and the distributive property to multiply a two-di...	4.NBT.B.5
FORWARD 2	Multiply two-digit numbers with partial products and the standard...	5.NBT.B.5

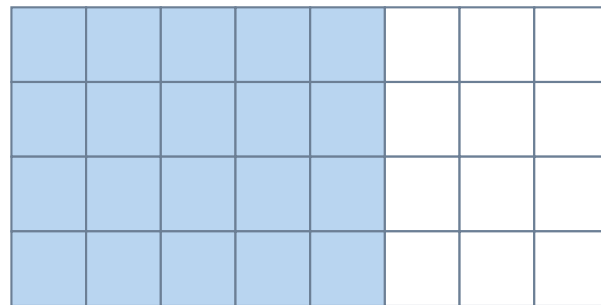
SET 53 • 3.MD.C.7.c

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.c

The 4-by-8 rectangle is split into a 4-by-5 part (shaded) and a 4-by-3 part.

Write the area as a sum of two products. Then find the total area.



Sum of products: _____

Total area: _____

A 6-by-9 rectangle is split into a 6-by-4 part and a 6-by-5 part.

Which expression gives its area?

A. $4 \times 5 + 6$

B. $6 \times 4 + 6 \times 5$

C. $6 \times 4 \times 5$

D. $6 + 4 + 5$

True or false?

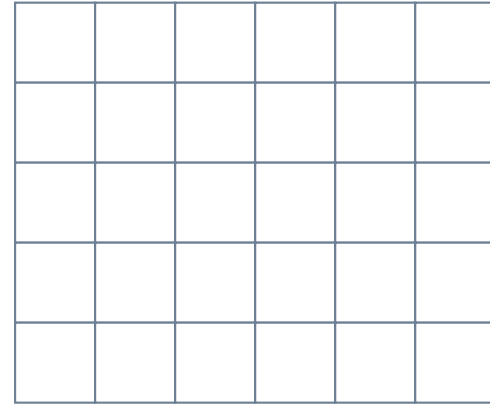
The area of a 3-by-7 rectangle equals $3 \times 5 + 3 \times 2$.

A. True

B. False

Draw one line along the grid to split the 5-by-6 rectangle into two smaller rectangles.

Write the total area as a sum of the two areas.



Sum: _____

True or false?

A 2-by-10 rectangle split into a 2-by-6 part and a 2-by-4 part has an area of $2 \times 6 \times 4$.

A. True

B. False

SET 53 • 3.MD.C.7.c

BACKWARD 1

Split an array into two parts and add • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

An array has 3 rows of 5 dots. A line splits it into 3 rows of 3 and 3 rows of 2.

How many dots are in each part? How many are there in all?

Parts: _____

Total: _____

True or false?

Splitting an array of 4 rows of 5 into 4 rows of 2 and 4 rows of 3 keeps 20 dots in all.

- A. True
- B. False

An array of 2 rows of 5 is split into 2 rows of 2 and 2 rows of 3.

How many dots are in the 2-rows-of-3 part?

A. 5

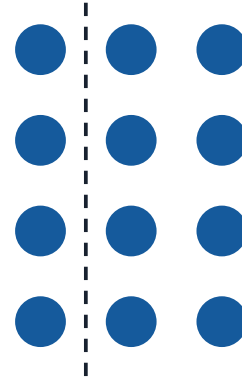
B. 4

C. 10

D. 6

The dashed line splits the array.

How many dots are there in all?



Dots: _____

True or false?

An array of 5 rows of 4 split into two parts has 10 dots in all.

- A.** True
- B.** False

SET 53 • 3.MD.C.7.c

BACKWARD 2

Add tens and ones within 100 • Grade 2 • Number & Operations in Base Ten •
2.NBT.B.5

Add.

$$20 + 15$$

Sum: _____

True or false?

$$30 + 24 = 54$$

A. True

B. False

$$40 + 32 = ?$$

A. 62

B. 8

C. 82

D. 72

Add.

$$18 + 12$$

Sum: _____

True or false?

$$35 + 25 = 50$$

A. True

B. False

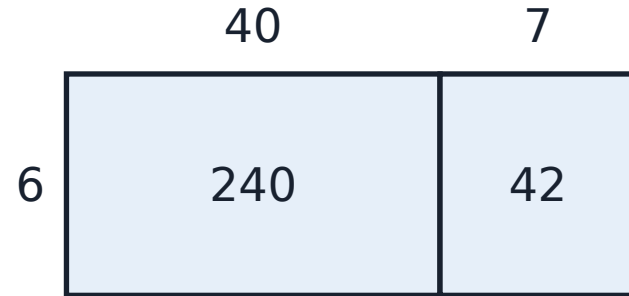
SET 53 • 3.MD.C.7.c

FORWARD 1

Use area models and the distributive property to multiply a two-digit number •
Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Use the area model to find
 6×47 .

Product: _____



True or false?

$$8 \times 53 = 8 \times 50 + 8 \times 3$$

A. True

B. False

Which two parts of an area model show 4×26 ?

- A.** 20×6 and 4
- B.** 4×2 and 4×6
- C.** 4×20 and 6
- D.** 4×20 and 4×6

Use an area model or the distributive property to find 7×38 . Show your work.

Product: _____

True or false?

$$5 \times 64 = 5 \times 60 + 4$$

A. True

B. False

SET 53 • 3.MD.C.7.c

FORWARD 2

Multiply two-digit numbers with partial products and the standard algorithm •

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

Use partial products to find 34×26 .

Product: _____

True or false?

$$25 \times 43 = 25 \times 40 + 25 \times 3$$

A. True

B. False

Which partial products give 12×35 ?

A. $12 \times 30 + 5$

B. $10 \times 35 + 2 \times 35$

C. $10 \times 35 + 2$

D. $10 \times 30 + 2 \times 5$

Use the standard algorithm to multiply. Show your work.

$$48 \times 27$$

Product: _____

True or false?

$$16 \times 25 = 16 \times 20 + 5$$

A. True

B. False

Find areas of rectilinear figures by splitting them into rectangles and adding the areas

MAIN	Grade 3 main question	3.MD.C.7.d
BACKWARD 1	Put shapes together to make a new shape	1.G.A.2
BACKWARD 2	Add two-digit numbers within 100	2.NBT.B.5
FORWARD 1	Find areas of figures made of rectangles in real-world problems	4.MD.A.3
FORWARD 2	Find the volume of a solid made of two prisms by adding their volu...	5.MD.C.5.c

SET 54 • 3.MD.C.7.d

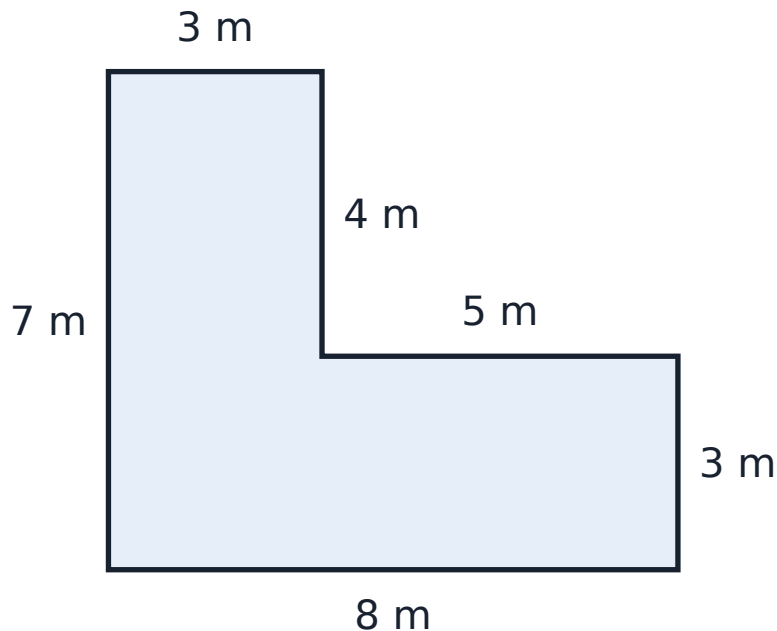
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.
Split it into two rectangles
and add their areas.

Areas of the parts: _____

Total area: _____



A figure is split into a 6-by-4 rectangle and a 2-by-3 rectangle that do not overlap.

What is the area of the figure?

- A.** 24 square units
- B.** 15 square units
- C.** 144 square units
- D.** 30 square units

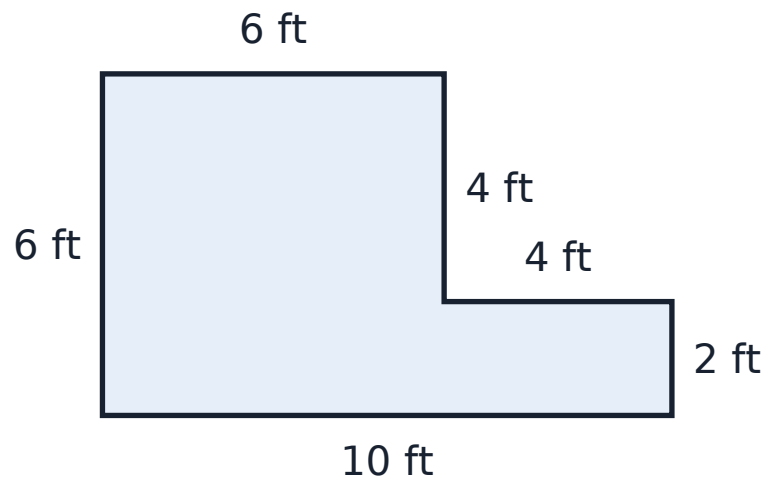
True or false?

The area of a figure made of two rectangles that do not overlap is the sum of the areas of the rectangles.

- A.** True
- B.** False

Find the area of the figure.

Area: _____

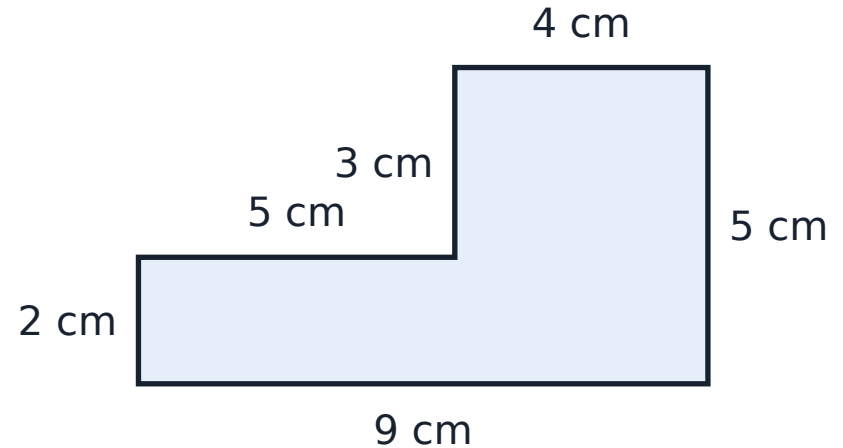


True or false?

The figure has an area of 50 square centimeters.

A. True

B. False



SET 54 • 3.MD.C.7.d

BACKWARD 1

Put shapes together to make a new shape • Grade 1 • Geometry • 1.G.A.2

True or false?

Two rectangles can be put together to make a new shape.

A. True

B. False

Two same-size squares are put side by side.
What shape do they make?

- A.** A triangle
- B.** A rectangle
- C.** A circle
- D.** A hexagon

How many same-size squares make a rectangle that is 3 squares long and 1 square wide?

Squares: _____

A shape is made of one square and one rectangle put together.

How many smaller shapes make it?

Shapes: _____

True or false?

Putting two rectangles together always makes a square.

A. True

B. False

SET 54 • 3.MD.C.7.d

BACKWARD 2

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$24 + 18$$

Sum: _____

True or false?

$$36 + 30 = 66$$

A. True

B. False

$$45 + 27 = ?$$

A. 62

B. 72

C. 18

D. 82

Add.

$$32 + 49$$

Sum: _____

True or false?

$$28 + 16 = 34$$

A. True

B. False

SET 54 • 3.MD.C.7.d

FORWARD 1

Find areas of figures made of rectangles in real-world problems • Grade 4 •
Measurement & Data • 4.MD.A.3

A room is made of a 12-ft by 10-ft rectangle and a 5-ft by 4-ft rectangle.

What is its total area?

Area: _____

True or false?

A garden made of a 15-m by 6-m bed and a 6-m by 6-m bed has an area of 126 square meters.

A. True

B. False

A deck is a 20-ft by 8-ft rectangle with a 6-ft by 5-ft hole cut out for a tree.

What is the area of the deck?

- A.** 160 square feet
- B.** 30 square feet
- C.** 130 square feet
- D.** 190 square feet

A parking lot is made of a 30-m by 15-m part and a 10-m by 15-m part.

What is its area?

Area: _____

True or false?

A figure made of a 9-by-7 rectangle and a 3-by-4 rectangle has an area of 23 square units.

- A. True
- B. False

FORWARD 2

Find the volume of a solid made of two prisms by adding their volumes • Grade 5 • Measurement & Data • 5.MD.C.5.c

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

Two prisms that do not overlap have volumes of 20 and 18 cubic units.

What is their total volume?

Volume: _____

True or false?

A 2-by-3-by-4 prism and a 2-by-3-by-1 prism together have a volume of 30 cubic units.

- A. True
- B. False

A solid is a 4-by-2-by-3 prism joined to a 2-by-2-by-3 prism.

What is its volume?

- A.** 24 cubic units
- B.** 36 cubic units
- C.** 288 cubic units
- D.** 12 cubic units

A building is a 10-by-6-by-3 prism with a 4-by-6-by-3 prism attached.

What is its total volume?

Volume: _____

True or false?

The volume of two joined prisms is the product of their volumes.

- A. True
- B. False

Find the perimeter of a polygon from its side lengths

MAIN	Grade 3 main question	3.MD.D.8
BACKWARD 1	Add lengths in word problems within 100	2.MD.B.5
BACKWARD 2	Add up to four two-digit numbers	2.NBT.B.6
FORWARD 1	Apply the perimeter formula for rectangles	4.MD.A.3
FORWARD 2	Find perimeters of shapes with fractional side lengths	5.NF.A.2

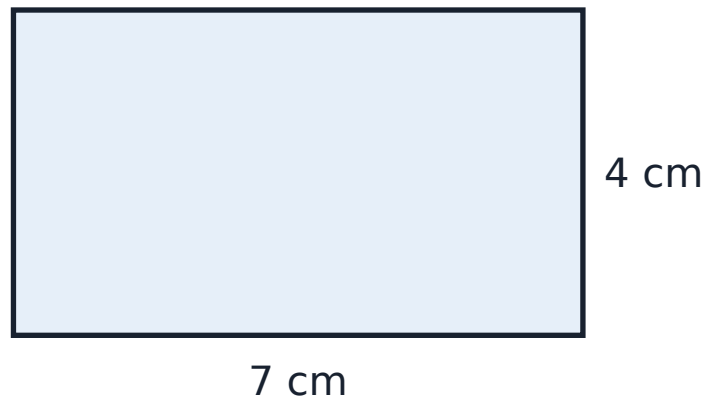
SET 55 • 3.MD.D.8

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



A triangle has sides of 6 in., 8 in., and 10 in.
What is its perimeter?

Perimeter: _____

A square has sides that are 9 m long.

What is its perimeter?

A. 81 m

B. 18 m

C. 27 m

D. 36 m

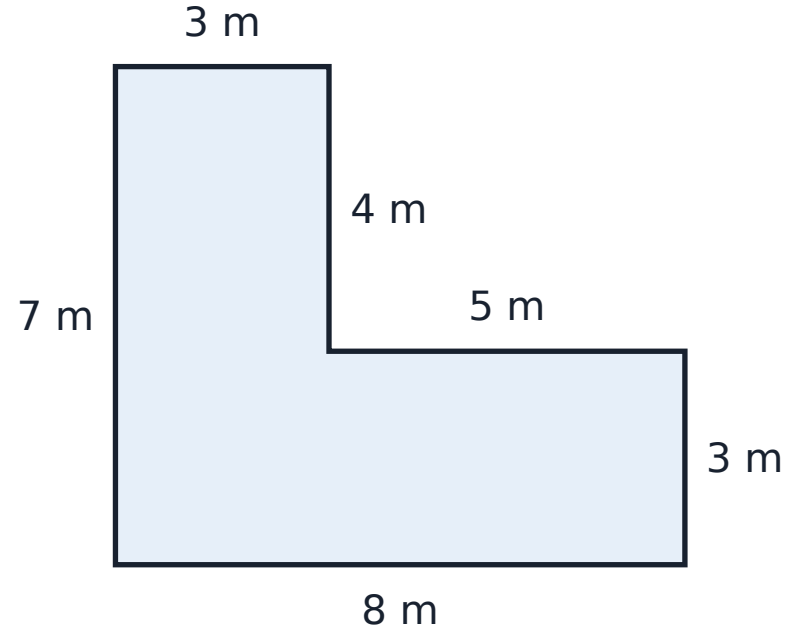
True or false?

A pentagon with five sides that are each 5 ft long has a perimeter of 25 ft.

- A. True
- B. False

Find the perimeter of the figure.

Perimeter: _____



SET 55 • 3.MD.D.8

BACKWARD 1

Add lengths in word problems within 100 • Grade 2 • Measurement & Data •
2.MD.B.5

A fence has parts that are 12 ft, 15 ft, and 20 ft long.
What is the total length?

Length: _____

True or false?

Ribbons of 18 cm and 25 cm have a total length of 43 cm.

A. True

B. False

Three sticks are 9 in., 14 in., and 21 in. long.

What is their total length?

- A.** 44 inches
- B.** 54 inches
- C.** 34 inches
- D.** 23 inches

A path has two parts, 35 m and 46 m long.

How long is the path?

Length: _____

True or false?

Boards of 30 in., 30 in., and 15 in. have a total length of 65 in.

A. True

B. False

SET 55 • 3.MD.D.8

BACKWARD 2

Add up to four two-digit numbers • Grade 2 • Number & Operations in Base Ten
• 2.NBT.B.6

Add.

$$12 + 15 + 12 + 15$$

Sum: _____

True or false?

$$20 + 8 + 20 + 8 = 56$$

A. True

B. False

$$9 + 16 + 9 + 16 = ?$$

A. 40

B. 41

C. 25

D. 50

Add.

$$30 + 25 + 18$$

Sum: _____

True or false?

$$11 + 11 + 11 + 11 = 40$$

A. True

B. False

SET 55 • 3.MD.D.8

FORWARD 1

Apply the perimeter formula for rectangles • Grade 4 • Measurement & Data •
4.MD.A.3

Use $P = 2 \times l + 2 \times w$ to find the perimeter of a 15-m by 9-m rectangle.

Perimeter: _____

True or false?

A 25-ft by 10-ft rectangle has a perimeter of 70 ft.

A. True

B. False

A square field has sides that are 45 yd long.

What is its perimeter?

- A.** 135 yd
- B.** 2,025 yd
- C.** 180 yd
- D.** 90 yd

A rectangle is 36 cm long and 14 cm wide.
What is its perimeter?

Perimeter: _____

True or false?

A 30-m by 20-m rectangle has a perimeter of 50 m.

A. True

B. False

SET 55 • 3.MD.D.8

FORWARD 2

Find perimeters of shapes with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A rectangle is $\frac{3}{4}$ ft by $\frac{1}{2}$ ft.

What is its perimeter?

Perimeter: _____

True or false?

A triangle with sides of $\frac{1}{2}$ m, $\frac{1}{3}$ m, and $\frac{1}{6}$ m has a perimeter of 1 m.

- A. True
- B. False

A square has sides that are $\frac{5}{8}$ inch long.

What is its perimeter?

A. $1\frac{1}{4}$ inches

B. $\frac{5}{32}$ inch

C. $2\frac{1}{2}$ inches

D. $\frac{20}{32}$ inch

A rectangle is $2\frac{1}{4}$ yd by $1\frac{1}{2}$ yd.

What is its perimeter?

Perimeter: _____

True or false?

A rectangle $\frac{2}{3}$ m by $\frac{1}{6}$ m has a perimeter of $\frac{5}{6}$ m.

A. True

B. False

Find an unknown side length of a polygon from its perimeter

MAIN	Grade 3 main question	3.MD.D.8
BACKWARD 1	Find an unknown part in a word problem within 100	2.OA.A.1
BACKWARD 2	Find the unknown number in an addition equation within 20	1.OA.D.8
FORWARD 1	Find an unknown side length from the perimeter of a rectangle wi...	4.MD.A.3
FORWARD 2	Find an unknown side length from a perimeter with fractions	5.NF.A.2

SET 56 • 3.MD.D.8

MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.D.8

A rectangle has a perimeter of 20 cm. Its length is 6 cm.
What is its width?

Width: _____

A triangle has a perimeter of 30 in. Two of its sides are 9 in. and 12 in.

How long is the third side?

Length: _____

A square has a perimeter of 28 m.

How long is each side?

- A.** 24 m
- B.** 112 m
- C.** 7 m
- D.** 14 m

True or false?

A rectangle with a perimeter of 18 ft and a length of 5 ft has a width of 8 ft.

A. True

B. False

A quadrilateral has a perimeter of 40 cm. Three of its sides are 10 cm, 12 cm, and 8 cm.

How long is the fourth side?

Length: _____

SET 56 • 3.MD.D.8

BACKWARD 1

Find an unknown part in a word problem within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A fence will be 50 m long. 32 m are built.

How many meters are left to build?

Meters: _____

True or false?

Kim needs 45 beads. She has 27 beads. She needs 18 more.

A. True

B. False

A path is 80 ft long. Lu walked some of it and has 35 ft left.

How far did she walk?

A. 35 ft

B. 45 ft

C. 55 ft

D. 115 ft

A lot has some cars and 14 trucks. There are 40 vehicles in all.

How many cars are there?

Cars: _____

True or false?

A tape is 60 cm long. 25 cm are used, so 45 cm are left.

A. True

B. False

SET 56 • 3.MD.D.8

BACKWARD 2

Find the unknown number in an addition equation within 20 • Grade 1 •
Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

$$12 + ? = 20$$

Unknown: _____

True or false?

In $? + 9 = 15$, the unknown number is 6.

A. True

B. False

$$18 = 10 + ?$$

- A.** 9
- B.** 8
- C.** 10
- D.** 28

Find the unknown number.

$$? + 7 = 19$$

Unknown: _____

True or false?

In $11 + ? = 17$, the unknown number is 5.

A. True

B. False

SET 56 • 3.MD.D.8

FORWARD 1

Find an unknown side length from the perimeter of a rectangle with the formula

• Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has a perimeter of 64 m and a length of 20 m.

What is its width?

Width: _____

True or false?

A square with a perimeter of 96 in. has sides that are 24 in. long.

A. True

B. False

A rectangle has a perimeter of 50 ft and a width of 9 ft.
What is its length?

- A.** 32 ft
- B.** 25 ft
- C.** 41 ft
- D.** 16 ft

A rectangular garden has a perimeter of 100 yd. Its length is 30 yd.

Write an equation with a letter for the width. Then solve it.

Equation: _____

Width: _____

True or false?

A rectangle with a perimeter of 44 cm and a length of 15 cm has a width of 14 cm.

- A. True
- B. False

SET 56 • 3.MD.D.8

FORWARD 2

Find an unknown side length from a perimeter with fractions • Grade 5 •
Number & Operations–Fractions • 5.NF.A.2

A triangle has a perimeter of 3 ft. Two of its sides are $\frac{3}{4}$ ft and $1\frac{1}{4}$ ft.

How long is the third side?

Length: _____

True or false?

A square with a perimeter of 5 m has sides that are $1\frac{1}{4}$ m long.

- A. True
- B. False

A rectangle has a perimeter of 6 in. and a length of $1\frac{3}{4}$ in.

What is its width?

A. $4\frac{1}{4}$ in.

B. $3\frac{1}{4}$ in.

C. $1\frac{1}{4}$ in.

D. $2\frac{1}{2}$ in.

A rectangle has a perimeter of 4 yd and a width of $\frac{2}{3}$ yd.

What is its length?

Length: _____

True or false?

A square with a perimeter of 2 m has sides that are $\frac{1}{4}$ m long.

- A. True
- B. False

Show rectangles with the same perimeter and different areas, or the same area and different perimeters

MAIN	Grade 3 main question	3.MD.D.8
BACKWARD 1	Recognize and draw rectangles	2.G.A.1
BACKWARD 2	Arrays with the same total	2.OA.C.4
FORWARD 1	Compare the areas and perimeters of rectangles with the formulas	4.MD.A.3
FORWARD 2	Find different prisms with the same volume — nearest related (no...	5.MD.C.5.b

SET 57 • 3.MD.D.8

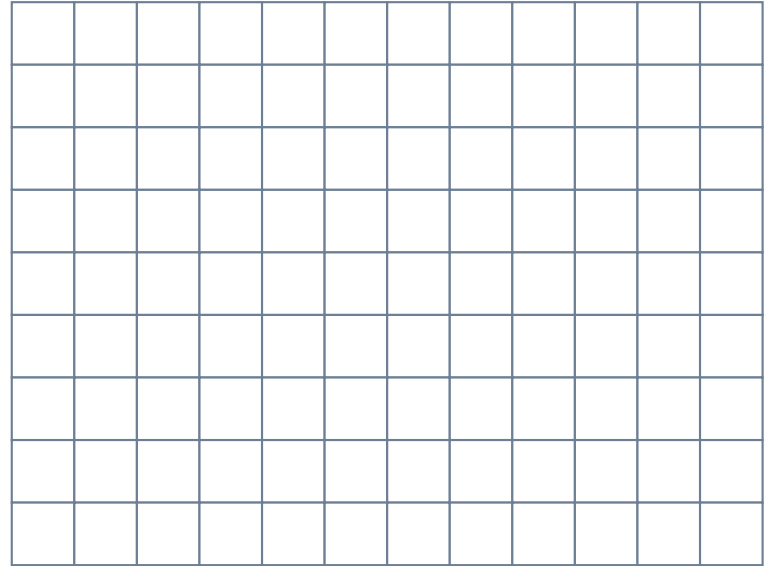
MAIN

Grade 3 main question • Grade 3 • Measurement & Data • 3.MD.D.8

Draw two different rectangles on the grid that each have a perimeter of 12 units.

What is the area of each rectangle?

Areas: _____



Which rectangle has the same area as a 4-by-6 rectangle but a different perimeter?

A. 2 by 10

B. 3 by 8

C. 4 by 5

D. 5 by 5

True or false?

A 2-by-6 rectangle and a 3-by-4 rectangle have the same area but different perimeters.

A. True

B. False

A 1-by-8 rectangle and a 2-by-4 rectangle both have an area of 8 square units.

Find the perimeter of each.

1 by 8: _____

2 by 4: _____

True or false?

Two rectangles with the same perimeter always have the same area.

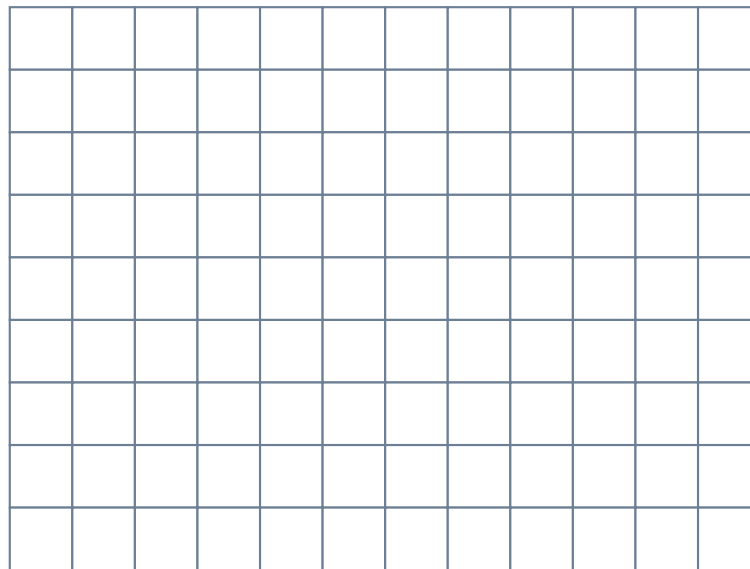
- A. True
- B. False

SET 57 • 3.MD.D.8

BACKWARD 1

Recognize and draw rectangles • Grade 2 • Geometry • 2.G.A.1

Draw a rectangle on the grid.



True or false?

A rectangle has 4 sides and 4 corners.

A. True

B. False

Which shape has 4 sides, with opposite sides the same length, and 4 square corners?

- A.** Triangle
- B.** Rectangle
- C.** Hexagon
- D.** Pentagon

How many sides does a rectangle have?

Sides: _____

True or false?

A rectangle has 3 corners.

A. True

B. False

SET 57 • 3.MD.D.8

BACKWARD 2

Arrays with the same total • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 2 rows of 6 dots. Another array has 3 rows of 4 dots.

How many dots does each array have?

2 rows of 6: _____

3 rows of 4: _____

True or false?

An array of 4 rows of 5 and an array of 5 rows of 4 have the same number of dots.

- A.** True
- B.** False

Which array has 10 dots?

- A.** 2 rows of 5
- B.** 3 rows of 3
- C.** 4 rows of 3
- D.** 2 rows of 4

How many dots are in 3 rows of 5?

Dots: _____

True or false?

An array of 2 rows of 3 and an array of 3 rows of 3 have the same number of dots.

- A.** True
- B.** False

SET 57 • 3.MD.D.8

FORWARD 1

Compare the areas and perimeters of rectangles with the formulas • Grade 4 •
Measurement & Data • 4.MD.A.3

A rectangle is 10 m by 6 m.

Find its area and its perimeter.

Area: _____

Perimeter: _____

True or false?

A 9-by-4 rectangle and a 12-by-3 rectangle have the same area.

- A. True
- B. False

Which rectangle has a perimeter of 30 ft and an area of 50 square feet?

- A.** 15 ft by 2 ft
- B.** 25 ft by 2 ft
- C.** 6 ft by 9 ft
- D.** 10 ft by 5 ft

A 2-by-10 rectangle and a 6-by-6 square both have a perimeter of 24 cm.

Which has the greater area?

Answer: _____

True or false?

A 7-by-7 square and an 8-by-6 rectangle have the same perimeter and the same area.

A. True

B. False

FORWARD 2

Find different prisms with the same volume • Grade 5 • Measurement & Data •
5.MD.C.5.b

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

Give the dimensions of two different prisms that each have a volume of 24 cubic units.

Prism 1: _____

Prism 2: _____

True or false?

A 2-by-2-by-6 prism and a 3-by-4-by-2 prism have the same volume.

- A. True
- B. False

Which prism has a volume of 36 cubic units?

A. 6 by 6 by 2

B. 2 by 2 by 8

C. 3 by 4 by 4

D. 3 by 3 by 4

A box is 5 by 2 by 3. Another box is 3 by 2 by ? and has the same volume.

What is the missing dimension?

Missing dimension: _____

True or false?

A 1-by-1-by-10 prism and a 2-by-2-by-2 prism have the same volume.

- A. True
- B. False

Understand that shapes in different categories can share attributes, and that quadrilaterals form a category

MAIN	Grade 3 main question	3.G.A.1
BACKWARD 1	Name triangles, quadrilaterals, pentagons, and hexagons	2.G.A.1
BACKWARD 2	Tell defining attributes from other attributes	1.G.A.1
FORWARD 1	Classify quadrilaterals by parallel sides and right angles	4.G.A.2
FORWARD 2	Attributes of a category belong to all of its subcategories	5.G.B.3

SET 58 • 3.G.A.1

MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.1

What do a rhombus, a rectangle, and a square always have in common?

- A.** They all have 4 sides.
- B.** They all have 4 square corners.
- C.** They all have 4 sides of equal length.
- D.** They all have 3 sides.

True or false?

A trapezoid and a square are both quadrilaterals.

A. True

B. False

A closed shape has 4 straight sides and 4 angles.
What category does it belong to?

Category: _____

Name an attribute that every rectangle and every square share, and an attribute that a square must have but a rectangle might not.

Shared: _____

Only the square must have: _____

True or false?

A triangle is a quadrilateral because it has straight sides.

A. True

B. False

SET 58 • 3.G.A.1

BACKWARD 1

Name triangles, quadrilaterals, pentagons, and hexagons • Grade 2 •
Geometry • 2.G.A.1

How many sides does a hexagon have?

Sides: _____

True or false?

A quadrilateral has 4 sides.

A. True

B. False

Which shape has 5 angles?

- A.** Triangle
- B.** Quadrilateral
- C.** Hexagon
- D.** Pentagon

What is a closed shape with 3 straight sides called?

Name: _____

True or false?

A pentagon has 6 sides.

A. True

B. False

SET 58 • 3.G.A.1

BACKWARD 2

Tell defining attributes from other attributes • Grade 1 • Geometry • 1.G.A.1

True or false?

A square can be any color and still be a square.

A. True

B. False

Which is a defining attribute of a triangle?

- A.** It has 3 sides.
- B.** It is red.
- C.** It is big.
- D.** It points up.

Is size a defining attribute of a rectangle?

Answer: _____

A closed shape has 4 straight sides.

Name one shape it could be.

Shape: _____

True or false?

A shape must be blue to be a hexagon.

A. True

B. False

SET 58 • 3.G.A.1

FORWARD 1

Classify quadrilaterals by parallel sides and right angles • Grade 4 • Geometry
• 4.G.A.2

A quadrilateral has exactly one pair of parallel sides.
What is it called?

Name: _____

True or false?

A rectangle has two pairs of parallel sides.

A. True

B. False

Which shape must have four right angles?

- A.** Rhombus
- B.** Parallelogram
- C.** Square
- D.** Trapezoid

Name a quadrilateral that has two pairs of parallel sides and no right angles.

Shape: _____

True or false?

Every quadrilateral has at least one pair of parallel sides.

- A. True
- B. False

SET 58 • 3.G.A.1

FORWARD 2

Attributes of a category belong to all of its subcategories • Grade 5 • Geometry
• 5.G.B.3

True or false?

All rectangles have 4 right angles, and every square is a rectangle, so every square has 4 right angles.

- A.** True
- B.** False

A shape is a square.

Which statement must be true?

- A.** It has 5 sides.
- B.** It is a rectangle.
- C.** It has exactly one pair of parallel sides.
- D.** It is a triangle.

Every parallelogram has opposite sides of equal length.
A rhombus is a parallelogram.

What must be true about the opposite sides of a rhombus?

Answer: _____

Does every rectangle have 4 sides? Explain using categories.

Answer: _____

Explanation: _____

True or false?

Every rhombus is a square.

A. True

B. False

Recognize rhombuses, rectangles, and squares as quadrilaterals, and draw quadrilaterals that are not in these subcategories

MAIN	Grade 3 main question	3.G.A.1
BACKWARD 1	Draw shapes with a given number of sides or angles	2.G.A.1
BACKWARD 2	Square corners and equal sides as defining attributes	1.G.A.1
FORWARD 1	Draw and identify parallel and perpendicular sides in quadrilatera...	4.G.A.1
FORWARD 2	Classify quadrilaterals in a hierarchy	5.G.B.4

SET 59 • 3.G.A.1

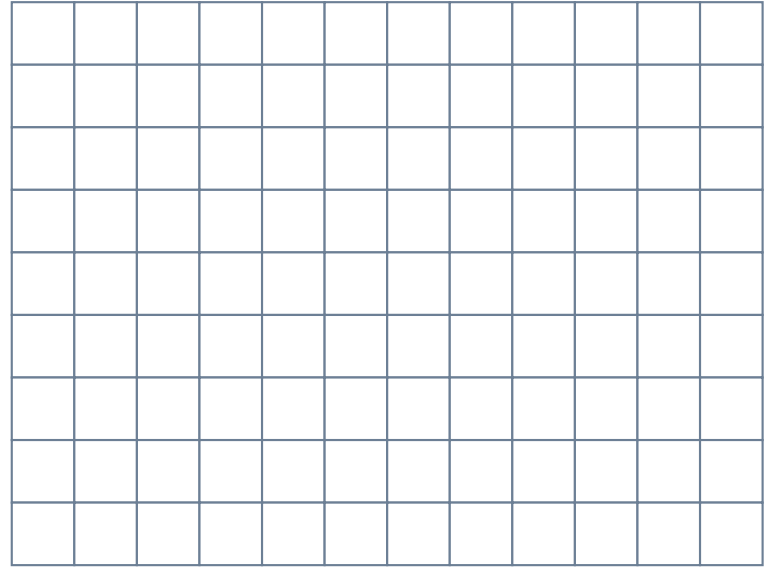
MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.1

Draw a quadrilateral that is not a rhombus, not a rectangle, and not a square.

What makes your shape a quadrilateral?

Reason: _____



Which shape is a quadrilateral but is NOT a rhombus, a rectangle, or a square?

- A.** A square
- B.** A rhombus
- C.** A rectangle whose length and width are different
- D.** A trapezoid with exactly one pair of parallel sides

True or false?

A square is a rectangle, and it is also a rhombus.

A. True

B. False

A shape has 4 sides of equal length and no square corners.

What is it?

A. A triangle

B. A rectangle whose length and width are different

C. A square

D. A rhombus that is not a square

True or false?

Every quadrilateral is a rectangle.

A. True

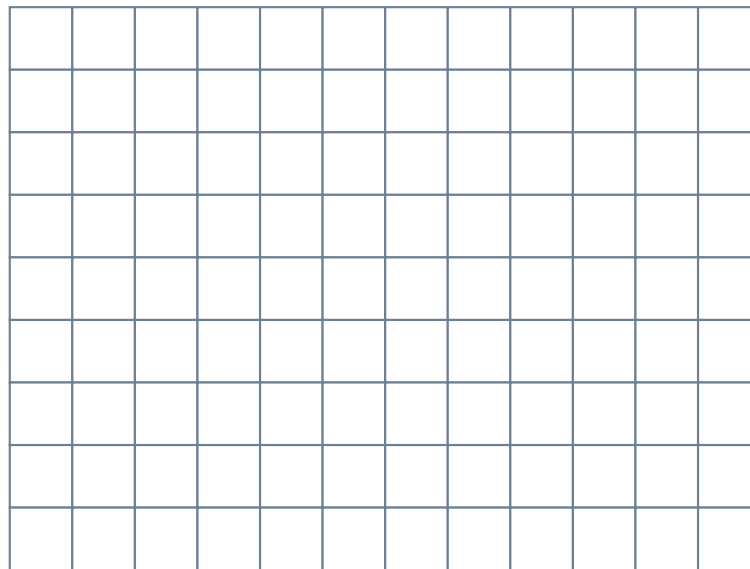
B. False

SET 59 • 3.G.A.1

BACKWARD 1

Draw shapes with a given number of sides or angles • Grade 2 • Geometry •
2.G.A.1

Draw a shape with 4 sides.



True or false?

A closed shape with 4 angles has 4 sides.

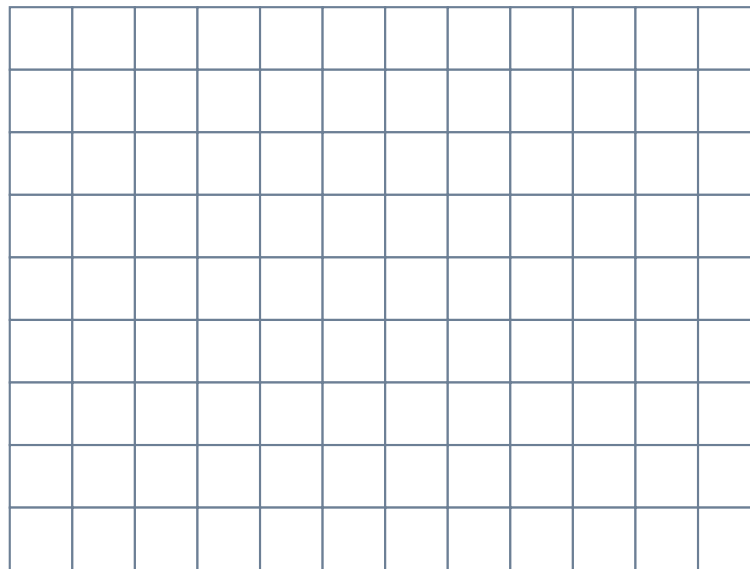
A. True

B. False

Which shape has 4 sides?

- A.** Triangle
- B.** Quadrilateral
- C.** Pentagon
- D.** Hexagon

Draw a shape with 3 angles.



True or false?

A closed shape with 5 sides has 4 angles.

A. True

B. False

SET 59 • 3.G.A.1

BACKWARD 2

Square corners and equal sides as defining attributes • Grade 1 • Geometry •
1.G.A.1

True or false?

A square has 4 sides of equal length.

A. True

B. False

How many square corners does a rectangle have?

Square corners: _____

Which shape always has 4 square corners?

- A.** Rectangle
- B.** Triangle
- C.** Hexagon
- D.** Circle

Is a rectangle that is turned on its corner still a rectangle?

Answer: _____

True or false?

A rectangle must have 4 sides of equal length.

A. True

B. False

SET 59 • 3.G.A.1

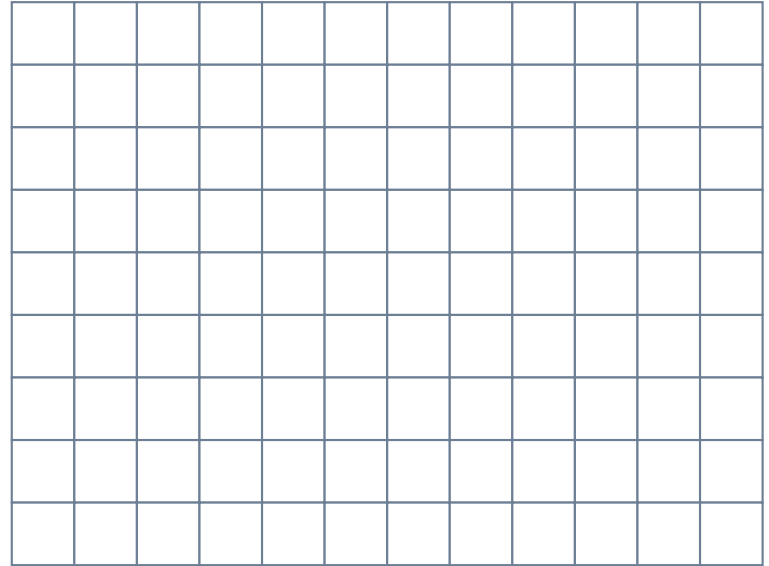
FORWARD 1

Draw and identify parallel and perpendicular sides in quadrilaterals • Grade 4 •
Geometry • 4.G.A.1

Draw a quadrilateral with exactly one pair of parallel sides.

What is it called?

Name: _____



True or false?

A rectangle has 4 right angles, so the sides that meet at each corner are perpendicular.

- A.** True
- B.** False

Which quadrilateral always has perpendicular sides?

- A.** A trapezoid with no right angles
- B.** A parallelogram with no right angles
- C.** A rectangle
- D.** A rhombus that is not a square

How many pairs of parallel sides does a square have?

Pairs: _____

True or false?

A rhombus that is not a square has perpendicular sides.

A. True

B. False

SET 59 • 3.G.A.1

FORWARD 2

Classify quadrilaterals in a hierarchy • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Square, rectangle, parallelogram, quadrilateral
- B.** Parallelogram, square, quadrilateral, rectangle
- C.** Quadrilateral, parallelogram, rectangle, square
- D.** Rectangle, quadrilateral, square, parallelogram

True or false?

Every square is a rhombus.

A. True

B. False

Which of these categories does every square belong to?
quadrilateral, parallelogram, rectangle, rhombus

Categories: _____

True or false?

Every rectangle is a square.

A. True

B. False

A parallelogram has 4 right angles and 4 sides of equal length.

What is its most specific name?

A. Rhombus

B. Quadrilateral

C. Rectangle

D. Square

Split shapes into parts with equal areas

MAIN	Grade 3 main question	3.G.A.2
BACKWARD 1	Split circles and rectangles into 2, 3, or 4 equal shares	2.G.A.3
BACKWARD 2	Draw rows and columns of same-size squares in a rectangle	2.G.A.2
FORWARD 1	Draw lines of symmetry that split a figure into matching parts —...	4.G.A.3
FORWARD 2	Tile a rectangle with unit-fraction squares or rectangles to find its...	5.NF.B.4.b

SET 60 • 3.G.A.2

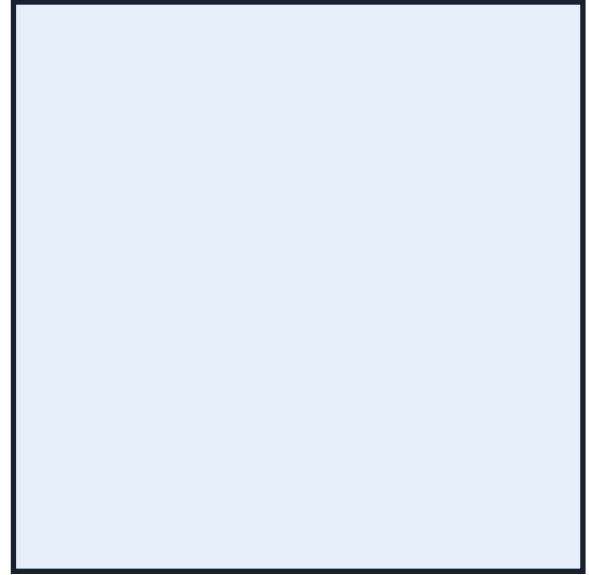
MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.2

Draw lines to split the rectangle into 4 parts with equal areas.



Draw lines to split the square into 3 parts with equal areas.



Which line always splits a rectangle into 2 parts with equal areas?

- A.** A line from a corner to the middle of the opposite side
- B.** Any line that crosses the rectangle
- C.** A line through the middle from one long side to the opposite long side, parallel to the short sides
- D.** A line close to one end

True or false?

One diagonal splits a square into 2 triangles with equal areas.

- A. True
- B. False

True or false?

If a rectangle is split into 4 parts, the parts always have equal areas.

- A. True
- B. False

SET 60 • 3.G.A.2

BACKWARD 1

Split circles and rectangles into 2, 3, or 4 equal shares • Grade 2 • Geometry •
2.G.A.3

Draw a line to split the rectangle into 2 equal shares.



True or false?

A rectangle can be split into 3 equal shares called thirds.

A. True

B. False

How many equal shares are there when a shape is split into halves?

A. 2

B. 3

C. 1

D. 4

Draw lines to split the square into 4 equal shares.



True or false?

Two shares of different sizes are halves.

A. True

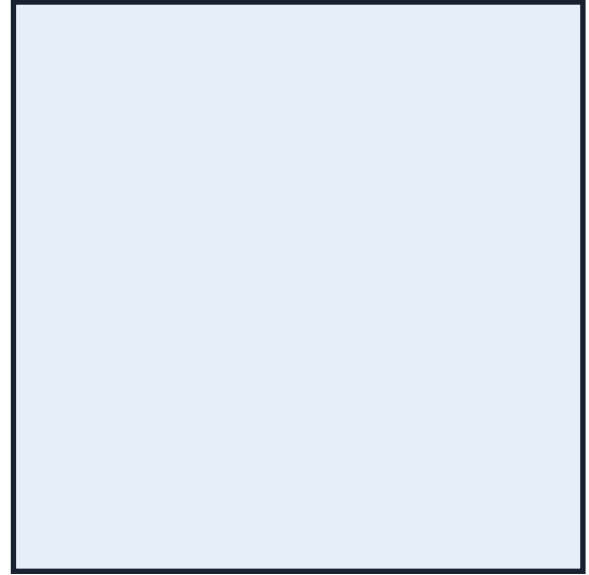
B. False

SET 60 • 3.G.A.2

BACKWARD 2

Draw rows and columns of same-size squares in a rectangle • Grade 2 •
Geometry • 2.G.A.2

Draw lines to split the square into 3 rows of 3 same-size squares.



True or false?

Splitting a square into 2 rows of 2 same-size squares makes 4 squares.

A. True

B. False

A rectangle is split into 1 row of 4 same-size squares.

How many squares are there?

A. 4

B. 8

C. 1

D. 5

A rectangle is split into 2 rows of 3 same-size squares.
How many squares are there?

Squares: _____

True or false?

A rectangle split into 3 rows of 2 same-size squares has 5 squares.

- A. True
- B. False

FORWARD 1

Draw lines of symmetry that split a figure into matching parts • Grade 4 •
Geometry • 4.G.A.3

► No Grade 4 standard directly continues this skill. This branch uses the nearest related Grade 4 standard.

Draw all the lines of symmetry of the rectangle.



True or false?

A line of symmetry splits a figure into two matching parts.

A. True

B. False

How many lines of symmetry does a square have?

A. 2

B. 1

C. 4

D. 0

How many lines of symmetry does a triangle with three sides of equal length have?

Lines: _____

True or false?

Every line that splits a rectangle into two parts with equal areas is a line of symmetry.

A. True

B. False

SET 60 • 3.G.A.2

FORWARD 2

Tile a rectangle with unit-fraction squares or rectangles to find its area • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.b

A square with an area of 1 square unit is split into 4 equal small squares.

What is the area of each small square?

Area: _____

True or false?

A rectangle $\frac{1}{2}$ unit by $\frac{1}{3}$ unit has an area of $\frac{1}{6}$ square unit.

- A. True
- B. False

A unit square is split into 3 equal columns and 2 equal rows.

What is the area of each small rectangle?

A. $\frac{1}{5}$ square unit

B. $\frac{1}{6}$ square unit

C. $\frac{1}{3}$ square unit

D. 6 square units

A rectangle $\frac{3}{4}$ unit by $\frac{1}{2}$ unit is tiled with rectangles that are $\frac{1}{4}$ unit by $\frac{1}{2}$ unit.

How many tiles fit? What is the area of the rectangle?

Tiles: _____

Area: _____

True or false?

A unit square split into 8 equal parts has parts with an area of $\frac{1}{4}$ square unit each.

- A. True
- B. False

Express the area of each equal part as a unit fraction of the whole

MAIN	Grade 3 main question	3.G.A.2
BACKWARD 1	Name one of 2, 3, or 4 equal shares	2.G.A.3
BACKWARD 2	Halves and quarters	1.G.A.3
FORWARD 1	Add fractions as joining parts of the same whole	4.NF.B.3.a
FORWARD 2	Add fractions with unlike denominators in word problems about pa...	5.NF.A.2

SET 61 • 3.G.A.2

MAIN

Grade 3 main question • Grade 3 • Geometry • 3.G.A.2

The rectangle is split into parts with equal areas.

What fraction of the whole area is each part?



Fraction: _____

A circle is split into 8 parts with equal areas.
What fraction of the area is one part?

A. $\frac{7}{8}$

B. $\frac{1}{7}$

C. $\frac{1}{8}$

D. $\frac{8}{1}$

True or false?

If a hexagon is split into 6 parts with equal areas, each part is $\frac{1}{6}$ of the area of the hexagon.

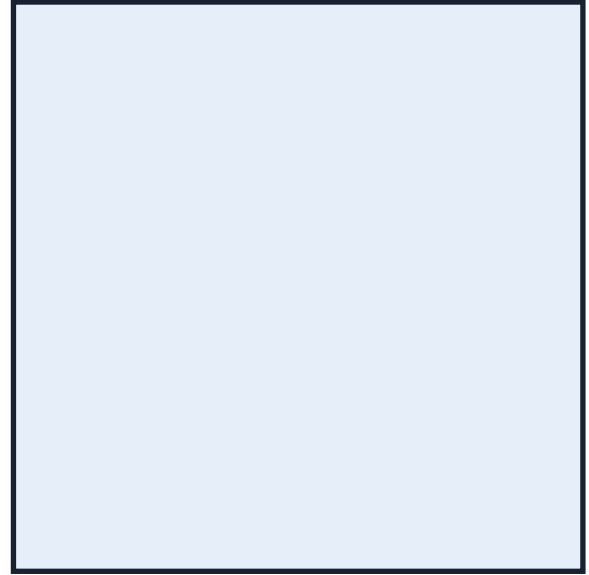
A. True

B. False

Split the square into 2 parts
with equal areas.

What fraction of the area of
the square is each part?

Fraction: _____



True or false?

A rectangle split into 3 parts with equal areas has parts that are each $\frac{1}{4}$ of the area.

- A. True
- B. False

SET 61 • 3.G.A.2

BACKWARD 1

Name one of 2, 3, or 4 equal shares • Grade 2 • Geometry • 2.G.A.3

A circle is split into 4 equal shares.

What is one share called?

Name: _____

True or false?

One of 2 equal shares is called a half.

A. True

B. False

A rectangle is split into 3 equal shares.

What is one share called?

- A.** A fourth
- B.** A third
- C.** A whole
- D.** A half

A shape is split into 2 equal shares.
What do the 2 shares make together?

Answer: _____

True or false?

One of 4 equal shares is called a third.

A. True

B. False

SET 61 • 3.G.A.2

BACKWARD 2

Halves and quarters • Grade 1 • Geometry • 1.G.A.3

A square is cut into 4 equal parts.

What is one part called?

Name: _____

True or false?

Two halves make a whole.

A. True

B. False

A pizza is cut into 2 equal parts.

What is each part called?

- A.** A quarter
- B.** A third
- C.** A half
- D.** A whole

How many halves are in one whole?

Halves: _____

True or false?

One of 4 equal parts is called a half.

A. True

B. False

SET 61 • 3.G.A.2

FORWARD 1

Add fractions as joining parts of the same whole • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

A rectangle is split into 6 equal parts. 2 parts are red and 3 parts are blue.

What fraction of the rectangle is red or blue?

Fraction: _____

True or false?

$\frac{2}{8} + \frac{3}{8}$ means joining 2 eighths and 3 eighths of the same whole.

A. True

B. False

A square is split into 4 equal parts. 1 part is shaded, and then 2 more parts are shaded.

Which equation shows the shaded fraction?

A. $\frac{1}{4} + \frac{2}{4} = \frac{2}{4}$

B. $\frac{1}{4} + \frac{2}{4} = \frac{3}{8}$

C. $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$

D. $\frac{1}{2} + \frac{2}{4} = \frac{3}{4}$

A pie is cut into 8 equal pieces. 3 pieces are eaten.
What fraction of the pie is left?

Fraction: _____

True or false?

$$\frac{1}{3} + \frac{1}{3} = \frac{2}{6}$$

A. True

B. False

SET 61 • 3.G.A.2

FORWARD 2

Add fractions with unlike denominators in word problems about parts of a whole • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

$\frac{1}{2}$ of a garden is corn and $\frac{1}{3}$ of it is beans.

What fraction of the garden is corn or beans?

Fraction: _____

True or false?

$\frac{1}{4}$ of a field is grass and $\frac{1}{2}$ is dirt. Together they are $\frac{3}{4}$ of the field.

A. True

B. False

A flag is $\frac{1}{3}$ red and $\frac{1}{6}$ white.

What fraction of the flag is red or white?

A. $\frac{1}{18}$

B. $\frac{1}{2}$

C. $\frac{2}{6}$

D. $\frac{2}{9}$

A cake is $\frac{3}{8}$ chocolate and $\frac{1}{4}$ vanilla.

What fraction of the cake is chocolate or vanilla?

Fraction: _____

True or false?

$\frac{1}{2}$ of a wall is blue and $\frac{1}{3}$ is green. The rest of the wall is $\frac{1}{5}$ of it.

A. True

B. False